

Ronit Sarkar

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[Portfolio](#) | [GitHub](#) | [LeetCode](#) | [GeeksforGeeks](#)

Education

Jadavpur University, BE in Computer Science and Engineering	August 2023 – June 2027
CGPA: 7.5 (Till 5th Semester)	
St. Paul's Academy, Burdwan	April 2020 - March 2023
Class XII (ISC): 85% Class X (ICSE): 96%	

Skills

Programming Languages: Java, C++, Python, C

Web Frameworks & Libraries NextJS, ReactJS, Express, FastAPI, Flask

Databases: PostgreSQL, MongoDB, Redis

Tools & Platforms: Git/GitHub, Linux, AWS

Relevant Coursework: Data Structures & Algorithms, Object Oriented Programming, Operating System, DBMS, Computer Networks, Computer Organisation & Architecture, Compiler Design

Projects

RedCache – Distributed In-Memory Key-Value Store | C++, Multithreading, TCP/IP [GitHub](#)

- Built a Redis-compatible in-memory data store from scratch, including a custom RESP protocol parser and multi-threaded TCP server supporting strings, lists, and hashes.
- Designed thread-safe data structures using `std::unordered_map`/`std::vector` synchronized with `std::mutex`, with hybrid passive/active TTL key eviction.
- Added automated snapshot persistence and crash recovery via a background disk-serialization thread and graceful shutdown handling.

Model Context Protocol (MCP) Server using Python | Python, FastMCP, Groq [GitHub](#)

- Built an async Model Context Protocol server (Python, FastMCP, uv) exposing web-scraping and search tools to LLMs over stdio.
- Integrated Serper API and Trafilatura to clean raw HTML from documentation sites, cutting LLM token usage and reducing hallucinations.
- Implemented a client-server RAG pipeline using `httpx` for concurrent requests and Llama-3.3-70B (Groq) for fast response synthesis.

GreenCart - Instant Grocery Delivery | React, Node.js, MongoDB, Stripe [GitHub](#) | [Website](#)

- Built a full-stack e-commerce platform (React 19, Node.js, Express, MongoDB) with a customer storefront and seller dashboard, securing sessions with HTTP-only JWT cookies and bcrypt hashing.
- Integrated Stripe Checkout with event-driven webhooks for real-time payment confirmation and Cloudinary for scalable image storage, automating order fulfillment tracking.
- Implemented persistent cart state, product filtering, and address management with Axios and React Router, deploying the decoupled frontend/API across Netlify and Vercel.

Achievements and Extracurricular

- Achieved **Rank 145 in the WBJEE 2023** among 100,000+ candidates.
- Selected as a **Google Campus Ambassador**, leading AI awareness and community engagement on campus.
- Represented Jadavpur University in **University Esports Championship 2025 National Finals**